

Program: Diploma in Artificial Intelligence	
Course Code: 6348	Course Title: R Programming Lab
Semester: 6	Credits: 2.5
Course Category: Program Core course	
Periods per week: 3 (L:1 T:0 P:3)	Periods per semester: 60

Course Objectives:

- Learn the techniques in R language.
- Familiar with data manipulation, analysis and visualization
- Understand control structures and R functions.

Course Prerequisites:

Topic	Course code	Course name	Semester
Sound knowledge of Python programming	1008	Introduction to IT systems Lab	I
Sound knowledge of Python programming	3347	Python Programming Lab	III
Knowledge of data sets	4347	Machine Learning Lab	IV

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Practice the basic programming concepts of R	9	Applying
CO2	Make use of the functions and data structures in R	12	Applying
CO3	Demonstrate file handling and data visualization in R	15	Applying
CO4	Open ended experiment	6	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Practice the basic programming concepts of R		
M1.01	Demonstrate the installation of R studio	1.5	Applying
M1.02	Develop basic programs using R	1.5	Applying
M1.03	Develop programs using strings	1.5	Applying
M1.04	Develop programs using Arrays	1.5	Applying
M1.05	Develop programs using R Objects: vectors, matrices, lists	3	Applying

CO2	Make use of the functions and data structures in R		
M2.01	Develop programs to implement functions in R	3	Applying
M2.02	Develop programs to implement factors in R	3	Applying
M2.03	Develop programs to create Data Frames	6	Applying
	Lab Exam – I	1.5	
CO3	Demonstrate file handling and data visualization in R		
M3.01	Illustrate the loading external dataset into R (csv, excel etc.)	3	Applying
M3.02	Develop an R program to create a data set and do statistical analysis on the data	6	Applying
M3.03	Develop programs to visualize data	6	Applying
CO4	Open ended experiment		
M4.01	Students work on an open-ended project using R to solve a real-world problem. Project might involve data analysis, visualization, and potentially some machine learning	15	Applying
	Lab Exam – II	1.5	

**** For example -Sentiment Analysis of Movie Reviews**

Text / Reference

T/R	Book Title/Author
R1	R for data science: Import, Tidy, Transform, Visualize, And Model Data Paperback – 20 January 2017 by Hadley Wickham (Author), Garrett Grolemund (Author)
R2	R For Dummies: How to Build Your Customer-Driven Growth Engine Paperback – 21 July 2015 by Andrie de Vries (Author), Joris Meys (Author)

Online Resources

Sl.No	Website Link
1	https://www.w3schools.com/r/
2	https://www.geeksforgeeks.org/r-tutorial/?ref=lbp